

Dita Pelaj

MSc Data Science Student | dita_s@live.com |



Computer Science graduate. MSc Data Science student with a strong background in Computer Science and hands-on experience in data analysis and full-stack development. Skilled at building reproducible pipelines, deploying web applications, and communicating technical concepts effectively.

EDUCATION

University of Potsdam

MSc. Data Science

Potsdam, Germany

Oct 2024 – Present

University of Pécs

BSc. Computer Science

Pécs, Hungary

2021 – 2024

- Stipendium Hungaricum Scholarship awardee
- Thesis topic: Educational Software for Formal Languages and Automata ([GenLang – Formal Languages & Automata Educational Software \(Python, React\)](#))
- GPA: 4.57 / 5.0

University of Prishtina

BSc. Mathematics (incomplete, two years completed)

Prishtina, Kosova

2017 – 2020

WORK EXPERIENCE

University of Potsdam – Software Engineering Institute

Research Assistant

Potsdam, Germany

Oct 2025 – Present

Tech Frame

Front-End Intern

Prishtina, Kosova

Jul 2023 – Aug 2023

- Developed dynamic user interfaces using HTML, CSS, and React, enhancing user experience and interactivity
- Gained a comprehensive understanding of WordPress and integrated its capabilities into web development projects.
- Connected WordPress APIs with React to enhance interactivity and responsiveness

Outdoor Kosova

Tour Guide

Radavc, Peja, Kosova

Jul 2017 – Aug 2021

- Summer job (June-September each year) in 'Sleeping Beauty' Cave in Radavc.
- Led groups of up to 30 people per tour inside the cave.

LANGUAGES

- Albanian: *Native*
- English: *Fluent*
- German: *Intermediate*
- Hungarian: *Basic*
- Norwegian: *Basic*
- Serbo-Croatian: *Basic*

SKILLS

- **Languages & Frameworks:** Python, Flask, HTML&CSS, Bootstrap, JavaScript, React, PHP, SQL, Java, SpringBoot, C++
- **Databases:** DB Modeling, MySQL, MSSQL Server, PostgreSQL
- **Tools:** Git, WordPress, Fly.io, Firebase, Jupyter, Snakemake

PROJECTS

AgroClimate – Climate & Crop Analysis (Python, Snakemake)

- Built a reproducible pipeline combining climate (DWD) and crop yield (OpenAgrar) data for Brandenburg.
- Applied statistical models (OLS, PCA) and visualizations to assess climate impacts and forecast yields.
- Automated workflows with Snakemake, generating 100+ plots and summaries for reproducible research.

Diagnosis Trends in German Hospitals (Python)

- Analyzed Destatis hospital diagnosis data (2014–2023) to uncover long-term trends and yearly patterns.
- Produced reproducible Jupyter notebooks with pandas, matplotlib, and seaborn for clear insights.

- Structured analysis with helper functions and documentation to ensure replicability.

GenLang – Formal Languages & Automata Educational Software (Python, React)

- Designed an educational web application to help students learn formal languages and automata through interactive grammar input and visualization.
- Implemented a Python (Flask) backend for grammar processing and a React frontend, ensuring smooth real-time interaction and feedback.
- Enhanced student engagement and usability by providing immediate language step-by-step generation outputs.
- Deployed a fully accessible web-based platform, making the tool available for broader student use and classroom integration.

Traffic Accident Analysis – Zurich (Python)

- Explored Zurich traffic accidents with visualizations (bar plots, heatmaps) and temporal analysis.
- Applied statistical and correlation methods to identify factors influencing accident frequency.

Games WebApp - Maze Game (JavaScript)

- Designed and implemented a maze game as part of a collaborative web app project.

MyWeatherApp (React)

- Built a weather app that retrieves real-time data via an external API for user-specified locations.

‘Online Courses’ Web API (Spring Boot)

- Developed a REST API for an online courses platform using Spring Boot and MySQL.
- Documented and tested endpoints with SwaggerUI to improve usability.

CERTIFICATES & COMPETITIONS

- PCEB webinar “Data Protection and Cyber Resilience” (June 2025)
- Simonyi International Hackathon - Pécs, Hungary (April 2023)
- Participation in StartHACK - Zürich, Switzerland (March 2023)
- Participation in K&H HACK3 Hackathon - Budapest, Hungary (February 2023)